Peer Response

by Thiago Contardi - Thursday, 14 August 2025, 3:55 PM

Your post gives a clear overview of how agent-based systems (ABS) have evolved and the benefits they bring to sectors like supply chain management and healthcare. While ABS can be robust in these scenarios, the accuracy and reliability of their outputs depend on the quality of the models and the data.

A model's predictions can be inaccurate, and its logic biased. To counter this, testing it against what has happened in the past is a fundamental step. We must also analyse how sensitive the results are to the assumptions we started with (Bonabeau, 2002). Without these checks, we cannot be sure the model's outputs align with the real world or provide a sound basis for making policy decisions.

In critical infrastructure contexts, like power grids, another preventive measure is the use of redundancy and fault-tolerant design at both the agent and communication levels. As Wooldridge (2009) notes, distributed systems must be resilient to agent or network failures, which can be achieved through fallback protocols, periodic health checks, and decentralised backup agents that can take over responsibilities.

These measures improve the reliability and enhance stakeholder's confidence, which is crucial when such systems are applied to high-stakes environments. By combining robust validation methods with fault-tolerant architecture, organisations can mitigate risks and maximise the value of agent-based solutions.

References

Bonabeau, E. (2002) Agent-based modeling: methods and techniques for simulating human systems. Proceedings of the National Academy of Sciences of the United States of America, 99 (Suppl. 3), pp. 7280–7287.

Wooldridge, M. (2009) An Introduction to MultiAgent Systems. 2nd edn. Chichester, UK: John Wiley & Sons.

Peer Response

by Rayyan Mohamed Abdalla Alshambeeli Alnaqbi - Sunday, 17 August 2025, 7:07 PM
Mansour, You gave a thorough and well-thought-out summary of how agent-based systems (ABS) have changed over time and why they are becoming more and more important in many fields. I really agree with your point that ABS lets people make decisions in real time and without a central authority. This is a big plus in today's fast-paced, data-driven world.
To add to your analysis, I want to point out that ABS also helps people be strong and flexible in times of crisis. For example, in systems for responding to disasters, agents can be given jobs like coordinating, searching, and rescuing. If someone fails to do their job or the situation changes, these agents can dynamically reassign tasks, which makes the system more robust (Chen et al., 2008). This ability to self-organize lets systems work well even when things go wrong or when there is a lot of stress.
Another big benefit is that agents can work together, which encourages new ideas and ongoing learning. In financial systems, agents can look at market trends, share their thoughts, and change their trading strategies based on what they see (Farmer and Foley, 2009). This means that ABS is both reactive and proactive, which is something that many traditional systems don't have.
Finally, I like that you brought up ABS in healthcare. In addition to modeling infectious diseases, ABS is now being used to plan personalized treatments where patient agents interact with medical protocols to create the best care pathways (Anwar et al., 2021).
In general, ABS has become a key tool for organizational intelligence, providing flexibility, fault tolerance, and predictive value in many areas.

References:

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Anwar, M., Abidi, S.R., Shepherd, M. and Abidi, S.S.R., 2021. Multi-agent systems for personalised healthcare: A systematic review. *Artificial Intelligence in Medicine*, 115, 102065. Available at: https://doi.org/10.1016/j.artmed.2021.102065 [Accessed 17 Aug. 2025].